Classification Lecture 10

Friday, 18 October 2024 6:16 AM

Q.1

Question: Which of the following statements is true regarding the handling of categorical and numerical data in classification models?

✗ A. Categorical data can be used directly in models without any preprocessing. → Decision Trees don't require preprocessing

✗ B. Numerical data must always be normalized or standardized before using in classification models.

✓ C. Categorical data often requires encoding techniques such as one-hot encoding or label encoding before it can be used in most classification models.

✗ D. Numerical data is less informative than categorical data and is typically excluded from classification models.

Q.2

Question: A binary classification model used in a financial fraud detection system generated the following confusion matrix from its test results:

	Predicted Positive (Fraud)	Predicted Negative (Not Fraud)
Actual Positive (Fraud)	120 TP	30 FN
Actual Negative (Not Fraud)	40 FP	310 TN

Calculate the following metrics based on the data provided in the confusion matrix. Select all the values that are correct (rounded to two decimal places):

✓ A. Precision for fraud detection: $\frac{120}{160}$
✓ B. Recall for fraud detection: $\frac{120}{150}$
✓ C. Accuracy of the model: $\frac{430}{500}$
✓ D. F1 Score for fraud detection: $2 \times \frac{(0.75\times0.80)}{(0.75+0.80)}$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{120}{160} = \frac{3}{4} = 0.75$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{120}{150} = \frac{4}{5} = 0.8$$

$$F_1 = \frac{2\times P\times R}{P+R} = 2\times\frac{0.75\times0.8}{(0.75+0.8)}$$

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} = \frac{120+310}{120+30+40+310}$$

$$= \frac{430}{500}$$

Q.3 Given a dataset of values [10, 20, 30, 40, 50], compute the z-score normalized value for the data point 30.

Sol.
$$z\text{-score} = \frac{x-\mu}{\sigma} \qquad \mu = \frac{10+20+30+40+50}{5} = 30$$

$$\sigma = \sqrt{\frac{(10-30)^2 + (20-30)^2 + \cdots + (50-30)^2}{5}}$$

$$z\text{-score}(30) = \frac{30-30}{\sigma} = \frac{0}{\sigma} = \boxed{0}$$

Q.4 **Question:** The softmax function is commonly used in classification models for multi-class classification tasks. Which of the following best describes the purpose of the softmax function in this context?

$$\frac{e^{z_i}}{\sum_{i=1}^{K} e^{z_i}}$$

× A. To convert the output scores from the model into binary values (0 or 1) for each class.

✓ B. To transform the model outputs into probabilities that sum up to 1, facilitating classification among multiple classes.

× C. To amplify the highest score output from the model to ensure that it is significantly higher than the others.

× D. To normalize the output scores to speed up learning by improving numerical stability.

Q.5

Question: Given the following data points in a 3-dimensional feature space, calculate the distances between them using different distance metrics. Select all correct distance values (rounded to two decimal places).

Data Points:

- Point A: $(2, 3, 4)$
- Point B: $(5, 7, 1)$
- Point C: $(1, 1, 1)$

Distance Metrics:

1. **Euclidean Distance**
2. **Manhattan Distance**
3. **Chebyshev Distance**
4. **Cosine Similarity**

Options:

✗ A. Distance between A and B (Euclidean) = 5.00
✗ B. Distance between A and C (Manhattan) = 5.00
✗ C. Distance between B and C (Chebyshev) = 4.00
✗ D. Distance between A and B (Cosine Similarity) = 0.32

Euclidean (A, B)

$$= \sqrt{(2-5)^2+(3-7)^2+(4-1)^2}$$

$$= \sqrt{3^2+4^2+3^2} = \sqrt{34}$$

Manhattan (A, C)

$$= |2-1| + |3-1| + |4-1|$$

$$= 1+2+3 = 6$$

chebyshev (B, C)

$$= \max(|5-1|, |7-1|, |1-1|)$$

$$= 6$$

$$\text{cosine}(A, B) = \frac{2\times5+3\times7+4\times1}{\sqrt{2^2+3^2+4^2}\ \sqrt{5^2+7^2+1^2}}$$

$$= \frac{10+21+4}{\sqrt{29}\cdot\sqrt{75}} \approx 0.751$$

Q.6

Question: Calculate the total cross entropy loss for the following three predictions made by a classification model. The model categorizes inputs into three classes: Red, Green, and Blue. Each row represents a different input with the predicted probabilities for each class and the actual class. Round your answer to three decimal places.

Data Point	Predicted: Red	Predicted: Green	Predicted: Blue	Actual Class
1	0.1	0.2	0.7	Blue ←
2	0.6	0.3	0.1	Red
3	0.2	0.7	0.1	Green

Sol.

$$\text{log loss} = \frac{1}{3}\left[-\log_e 0.7 - \log_e 0.6 - \log_e 0.7\right]$$

$$\approx \boxed{0.408}$$

Q.7

Question: Consider a multi-label classification system designed to categorize news articles into multiple categories including Politics, Economy, Technology, and Health. After processing a batch of articles, the following predictions and actual labels are observed for four different articles:

Article ID	Actual Labels	Predicted Labels	Predicted Ranking	
1	Politics, Economy	Economy, Technology	[E, T, P, H]	$1/4 = 0.25$
2	Economy, Technology	Economy, Technology, Health	[E, T, H, P]	0
3	Technology	Technology	[T, E, P, H]	0
4	Politics, Health	Economy, Health	[E, H, P, T]	$2/4 = 0.5$

Based on this data, calculate the following metrics and select the correct values (expressed in fractions):

✗ A. Hamming Loss = $\frac{5}{14}$

✓ B. Subset Accuracy = $\frac{1}{4}$

✗ C. Coverage Error = $\frac{10}{4}$

✗ D. Ranking Loss = $\frac{5}{14}$

$$\text{Hamming loss} = \frac{2+1+0+2}{4\times 4} = \frac{5}{16}$$

$$\text{Subset Accuracy} = \frac{1}{4}$$

$$\text{Coverage error} = \frac{3+2+1+3}{4} = \frac{9}{4}$$

$$\text{Ranking loss} = \frac{0.25+0.5}{4} = \frac{3}{16}$$

Q.8.

Question: Which of the following statements about the k-Nearest Neighbors (KNN) algorithm are true? Select all that apply.

✓ A. KNN is a supervised learning algorithm that can be used for both classification and regression tasks.

→ assign the average value instead of majority

✗ B. The value of k must always be an odd number to avoid ties in classification.

✓ C. KNN requires feature scaling (normalization or standardization) to ensure that all features contribute equally to the distance calculations.

→ because of distance calculations

✓ D. KNN is sensitive to irrelevant features, and including too many irrelevant features can negatively impact its performance.

Q.9.

Question: Consider the following dataset consisting of three points in a 2D feature space, classified into two categories: A (Circle) and B (Square).

Point ID	Feature 1	Feature 2	Class
1	1	1	A
2	2	2	A
3	3	3	B

Using the KNN algorithm with $k = 2$, determine the class of the following new point:

- **New Point:** (2.5, 2.5)

Based on the distances calculated, select all correct statements regarding the classification of the new point.

Options:

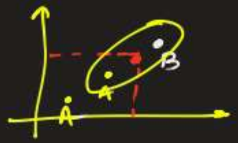
✗ A. The distance from the new point to Point 1 is approximately 1.41.

✓ B. The distance from the new point to Point 2 is approximately 0.71.

? C. The new point (2.5, 2.5) will be classified as Class A (Circle) based on majority voting among its 2 nearest neighbors.

↳ Depending on how we resolve ties.

✓ D. The distance from the new point to Point 3 is approximately 0.71.



Pt1:- $\sqrt{(2.5-1)^2+(2.5-1)^2}$

$= \sqrt{1.5^2+1.5^2}$

≈ 2.12

Pt2:- $\sqrt{(2.5-2)^2+(2.5-2)^2}$

$= \sqrt{0.5^2+0.5^2}$

$= \sqrt{0.5}$

≈ 0.71

Pt3:- ≈ 0.71

Q.10

Question: Which of the following statements accurately describes the Naive Bayes classifier? Select the correct option.

independent

× A. Naive Bayes assumes that the features are dependent on each other given the class label.

× B. Naive Bayes can only be used for binary classification problems.

✓ C. The classifier is based on Bayes' Theorem, which calculates the posterior probability of a class given the features.

× D. Naive Bayes performs better when the assumption of feature independence is violated.

Q-11

Question: Consider a dataset used for sentiment analysis to classify movie reviews as either Positive or Negative. The dataset contains the following features extracted from the reviews:

- Feature 1: Contains the word "great"
- Feature 2: Contains the word "boring"
- Feature 3: Contains the word "amazing"

After training a Naive Bayes classifier on this dataset, the following probabilities were calculated:

- Prior Probability:
  - P(Positive) = 0.6
  - P(Negative) = 0.4
- Likelihoods:
  - P(Feature 1 | Positive) = 0.8
  - P(Feature 1 | Negative) = 0.2
  - P(Feature 2 | Positive) = 0.1
  - P(Feature 2 | Negative) = 0.6
  - P(Feature 3 | Positive) = 0.9
  - P(Feature 3 | Negative) = 0.3

Given a new review that contains the words "great," "boring," and "amazing," calculate the posterior probabilities using the Naive Bayes classifier. Select all true statements based on your calculations.

Options:

✓ A. The posterior probability P(Positive | "great", "boring", "amazing") is greater than P(Negative | "great", "boring", "amazing").

✓ B. The Naive Bayes classifier assumes that the features are independent given the class label.

✓ C. The presence of the word "boring" decreases the likelihood of the review being classified as Positive.

✗ D. The final classification will favor the Negative class because it has a higher prior probability.

$$P(pos \mid \text{"g, b, a"}) \propto P(g \mid pos) \cdot P(b \mid pos) \cdot P(a \mid pos) \cdot P(pos)$$

$$= 0.8 \times \underline{0.1} \times 0.9 \times 0.6$$

$$= 0.0432$$

$$P(Neg \mid \text{"g, b, a"}) \propto 0.2 \times 0.6 \times 0.3 \times 0.4$$

$$= 0.0144$$

Q.12

Question: A Naive Bayes classifier is used for a multi-class classification problem with the following characteristics:

- Number of features: 8
- Each feature can take 4 different values.
- Number of classes: 5

Using the Naive Bayes assumption, how many parameters are needed for the model? Provide your answer as a single integer.

Sol. $p = 8, \quad m = 4, \quad K = 5$

$$\#par = Kp(m-1) + (K-1) = 5 \times 8 \times 3 + 4$$

$$= 120 + 4$$

$$= \boxed{124}$$

Q.13

Question: Which of the following statements about Logistic Regression in classification are true? Select all that apply.

✗ A. Logistic Regression can only be used for binary classification problems. → use softmax for multi-class classification

✓ B. The logistic function (sigmoid function) transforms the linear combination of inputs into a probability value between 0 and 1.

✓ C. In Logistic Regression, the coefficients represent the change in the log-odds of the dependent variable for a one-unit increase in the independent variable.

✓ D. Logistic Regression assumes a linear relationship between the independent variables and the probability of the dependent variable being 1.

↳ log (prob) log odds

Q.14

Question: A logistic regression model is trained to predict whether a student will pass (1) or fail (0) an exam based on their study hours and attendance. The final model parameters are as follows:

- Intercept (b_0): -1.5
- Coefficient for Study Hours (b_1): 0.4
- Coefficient for Attendance (b_2): 0.6

Given a new student with the following data:

- Study Hours: 5
- Attendance: 80% (0.8 when normalized)

Using the logistic regression model, calculate the following values and select all that are correct (rounded to four decimal places):

1. The linear combination z. ✓
2. The predicted probability $P(y=1)$ that the student will pass the exam. ✓
3. The decision threshold for classification, assuming the threshold is set at 0.5.

$$z = -1.5 + 0.4 \times 5 + 0.6 \times 0.8$$
$$= 0.98$$
$$P(y=1) = \sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0.98}}$$
$$\approx 0.727$$

Decision :- $y=1$

Options:

✗ A. The linear combination z = 0.5

✗ B. The predicted probability $P(y=1)$ = 0.6225

✗ C. The predicted probability $P(y=1)$ = 0.7311

✓ D. The student is classified as passing the exam.